The Geometry of Duration and the In-Between

Bergson — Heidegger/Zollikon —
Kimura/Kyōto — Corbin/Annales — GI Theory

Étale Cohomology

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持続と「あいだ」の幾何学

ベルクソン・ハイデガー＝ツォリコーン・木村敏＝京都学派・コルバン＝アナール学派——そして GI 理論

Die Geometrie der Dauer und des Zwischen

Bergson — Heidegger/Zollikon — Kimura/Kyōto — Corbin/Annales — GI-Theorie

La géométrie de la durée et de l’entre

Bergson — Heidegger/Zollikon — Kimura/Kyōto — Corbin/Annales — théorie GI

Étale Cohomology

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πάντα ῥεῖ καὶ οὐδὲν μένει.

"Everything flows and nothing abides."

「万物は流れ、何もとどまらない」

« Tout coule et rien ne demeure. »

„Alles fließt und nichts bleibt.“

— Hērakleitos, DK 22 B12

Preface

> *La durée toute pure est la forme que prend la succession de nos états de conscience quand notre moi se laisse vivre, quand il s'abstient d'établir une séparation entre l'état présent et les états antérieurs.*
>
> "Pure duration is the form taken by the succession of our conscious states when the ego lets itself live, when it refrains from establishing a separation between the present state and the anterior states."
>
> 「純粋持続とは、われわれの自我が生きるにまかせ、現在の状態と先行する諸状態とのあいだに分離を立てることを控えるとき、意識諸状態の継起がとる形式である」
>
> „Die reine Dauer ist die Form, welche die Aufeinanderfolge unserer Bewusstseinszustände annimmt, wenn unser Ich sich leben lässt, wenn es davon absieht, eine Trennung zwischen dem gegenwärtigen Zustand und den vorangegangenen Zuständen zu setzen."
>
> *— Henri Bergson, Essai sur les données immédiates de la conscience (1889), Chapter II*

As a young man, Nishida Kitarō had wished to become a mathematician. Having passed through the Fourth Higher School in Kanazawa, he entered the Imperial University of Tōkyō in the special division of philosophy—that is to say, as an irregular student, not a regular one; a circumstance he returned to repeatedly in later years, confessing that it had left him with the feeling of having become one of life's failures. And yet, beneath the entire surface of his philosophical meditation, there ran an unfailing current of mathematical passion, like groundwater that no drought can exhaust. Even in the final pages of his diary one finds the entry: "Today, again, I did mathematics." The Dedekind cut (Schnitt)—that cardinal notion of the foundations of mathematics which places the continuity of the real numbers on rigorous footing—and other mathematical structures he studied at the level of the professional mathematician, absorbing them into his own "logic of place" (basho no ronri).

It was during my years at secondary school that I first learnt of this fact, and it struck me with the force of a silent thunderbolt. That Nishida had aspired to be a mathematician, that even after turning to philosophy he had never let go of mathematics—that mathematics and the humanities could, within the intellectual enterprise of a single human being, fuse to the point of inseparability—this recognition constituted, for me, the first primal experience of the point at which mathematics and the human sciences touch. My encounter with the French language, however, lay still further back. At nursery school, my parents made me recite from memory the original text of Saint-Exupéry's Le Petit Prince. The meaning eluded me entirely. Yet the succession of French sounds—a system of resonances utterly foreign to the mother tongue—settled in the child's consciousness like a sediment that no forgetting would ever dissolve. It was perhaps the most primordial experience of what Bergson calls "pure duration": duration as a qualitative flow of sounds, prior to all signification. At the same period I fell under the spell of the philosophy textbooks for the French lycée and the examination papers of the baccalauréat. Their author was Paul Foulquié (1893–1983). A Jesuit and teacher of philosophy, Foulquié had published a Précis de philosophie à l'usage des candidats au baccalauréat (first edition 1936)—a classical two-volume textbook spanning psychology, logic, morals and metaphysics—which opened to a schoolboy in a distant country the gates of French intellectual culture. His name I

shall not forget. Surrounded by Irish, English, Americans and French, receiving from them my earliest instruction, I heard for the first time outside the classroom, as living words rather than printed names, the names of Descartes, Pascal and Bergson. If the present essay takes Bergson's pure duration as its point of departure, it is in this adolescent experience that its roots are to be found.

It was also at this time that I learnt that the Russians, too, carried within themselves the torment of not knowing whether they belonged to the lineage of Europe or to that of Asia—a crisis of civilisational identity. It was during a summer spent in the Lake District of northern England and in Switzerland. In 1836, when Pyotr Chaadayev (Пётр Яковлевич Чаадаев, 1794–1856) published his Philosophical Letters (Философические письма), declaring Russia a void on the margins of Western civilisation—Nicholas I had Chaadayev declared insane—there began the two-hundred-year quarrel between Slavophiles and Westernisers. Does Russia belong to Europe, or does it constitute a civilisation of its own? This question, which I shall have occasion to examine later through the case study of Peter the Great, I learnt not from books but from the living voices of Russians themselves. And I perceived, with something approaching astonishment, that this torment bore an uncanny resemblance to the anxiety the Japanese have carried within themselves since the Meiji Restoration: to learn the techniques and institutions of the West while searching for a place in which to locate one's own civilisational identity. The question that Chaadayev posed, that Peter put into practice, that the leaders of Meiji reformulated, and that Nishida Kitarō attempted to answer through his "logic of place"—the Russians, too, posed it without cease. Putyatin appeared before the Japanese earlier than Perry. In truth, the Russians never ceased to be, for the Japanese, closer than is commonly supposed.

To be invited for tea in an English household was for me a moment of unclouded happiness. Equally, the hours spent over the samovar in the homes of Russian friends possessed a warmth and tenderness entirely their own. From Turkish friends I learnt the depth of a world-view that conceives of itself as a bridge between West and East. Turkey, like Japan and Russia, took in the traditional spiritual world and the values of Western Europe at distinct strata—the stratum of culture and society on the one hand, that of political organisation and form of government on the other—but in a manner peculiar to itself. Harmony and friction. Consonance and dissonance. Most Japanese have not yet listened with sufficient attentiveness to the voice that Russia and Turkey—two remarkable civilisational worlds—ceaselessly address to Japan.

Edo, the capital of the Tokugawa shōgunate, was a city raised upon a plain in the Kantō that had been nothing but marshland until vast feats of engineering rendered it habitable. What Ieyasu accomplished after the Taikō Hideyoshi had banished him to these desolate swamps—a reshaping of the terrain itself, a remaking of the topography—corresponds, in the language of GI theory, to a Lie derivative whose value must have been considerably large. That a city should, in the early modern period, come to surpass Paris and London in population: this is an enterprise structurally isomorphic to that of Peter the Great raising Saint Petersburg upon the marshes at the mouth of the Neva. I sense, moreover, a structural analogy between the founding of Edo and the founding of the United States of America—those peoples who, leaving the European continent behind for the New World, built New Amsterdam—the land that would soon bear the

name New York—and brought a nation to flourishing in every dimension: economic, cultural, civilisational, political, technological and military. To connect the building of Edo and the establishment of the shōgunate—far from Nara, Kyōto, Kamakura and Muromachi—with the founding narrative of the United States far from England: that, I believe, is a perspective worth retaining.

It was at secondary school that I discovered the story of Nicolas Bourbaki. In 1934, André Weil, Henri Cartan, Jean Dieudonné, Claude Chevalley and a handful of other young French mathematicians, revolted by the ossification of mathematical education in France—where nineteenth-century textbooks were still in perpetual use and neither set theory nor topology was taught in any systematic fashion—resolved to rewrite the whole of mathematics from scratch. Under the fictitious name "Bourbaki" they inaugurated the publication of the Éléments de mathématique. When I learnt of this, the gentle illusion that had formed within me—"France is the republic of the mind"—nurtured by the admirable prose of successive Ministers of Culture, wavered for a moment. That the mathematical education of France, that supposed republic of the mind, had been so ossified as to require wholesale renovation at the hands of the Bourbaki salon: here was food for thought on the relation between institutions and intelligence. Yet the intellectual force of the Bourbakists, their passion, and above all the tenacity of a will sustained across decades of publication—an intellectual practice of Bergsonian "duration" if ever there was one—held my esteem for France fast and kept it from foundering.

Today, category theory (théorie des catégories) is gradually overgrowing the rational edifice that Bourbaki had erected. Bourbaki placed set theory at the foundation of mathematics; category theory substitutes the relations between objects—morphisms and functors—as the primary foundation. This reversal resonates structurally with the one effected by Kimura Bin when he accorded primacy to "koto" over "mono," to the in-between over substance. A new Bourbaki is therefore awaited—a collective capable of rewriting the whole of mathematics from the standpoint of the in-between, adequate to the categorical age. Does such a collective exist today, at the École normale supérieure or the École polytechnique? The question remains open.

It was at a time not far removed from that discovery that I first took up Eugène Minkowski's (1885–1972) Le Temps vécu: Études phénoménologiques et psychopathologiques (1933). Fifteen months later I was wholly immersed in the philosophy of the body of Merleau-Ponty (1908–1961). The Phénoménologie de la perception (1945), in describing the "lived body" (corps vécu)—refusing the Cartesian dualism of mind and body, conceiving the body as the very medium of our engagement in the world—does nothing other than concretise, at the plane of the flesh, the Heideggerian being-in-the-world, while furnishing at the same time the philosophical ground for the body as the place in which Kimura's "in-between" comes to dwell. The memory of riding—a child's body absorbing through legs and buttocks the vibrations of the horse—was linked, through the phenomenology of perception, to Kimura Bin's "in-between," many years later. For me this was not a coincidence: it was an underground necessity.

If, in the case studies of GI theory, I repeatedly ran on servers the computations that compared the behaviour of AI agents endowed with a body against those deprived of one, and if I set down the results with care, it was by virtue of a Merleau-Pontian inheritance whose hold on me has never slackened and never will. Body and self-consciousness, the life-world (Lebenswelt), Uexküll's Umwelt, being-in-the-world; and then the thought of the middle Heidegger—Die Frage nach dem Ding (1935–36), the question concerning technology (1953), the Beiträge zur Philosophie (Vom Ereignis, 1936–38), culminating in the Fourfold (das Geviert), that conjunction of sky and earth, divinities and mortals whose experience no concept can contain: all of this I projected onto the AI agents, hoping to discover in the geometry of decision-making some trace of the ontological difference. An agent possessed of a body forms with its environment an "in-between"; an agent without a body lacks it. To quantify, by means of the Lie derivative and scalar curvature, the geometric trace that this difference imprints upon the agent's decision-making structure: therein lay the core of the case studies in Volume 2 of GI theory.

The intellectual legacy of Merleau-Ponty's philosophy of the body must at all costs be brought into AI agent research. That present-day AI research conceives of embodiment as nothing more than a "sensor-data input channel" is the unconscious repetition of Cartesian dualism—exactly the same error that Merleau-Ponty had denounced seventy years earlier. To inscribe in the design principles of the AI agent the body as mediation of engagement in the world—the "lived body"—and to constitute the appropriate academic discursive space: this appeared to me as one of my obligations.

Heidegger's "Building Dwelling Thinking" (Bauen Wohnen Denken, 1951) is a text of lived praxis that brings together architectural space and bodily space, being-in-the-world and the Fourfold (das Geviert). So at any rate I received it. To dwell (Wohnen) is to sojourn under the Fourfold—sky and earth, the divinities and the mortals; to build (Bauen) is to give form to the place of that sojourn. Not the technical act of construction, but the ontological meditation on dwelling. I later discovered that a group of existential architects had taken up this Heideggerian thought in architectural theory. In Japan several translations were published by Kashima Publishing. Genius Loci: Towards a Phenomenology of Architecture by Christian Norberg-Schulz (1926–2000) is the representative work.

The organic architecture of Frank Lloyd Wright (1867–1959) had long commanded my attention, for here was an architecture whose governing principle lay not in abolishing the "in-between" that separates nature from building but in orchestrating their mutual interpenetration. I saw in it the realisation of the Heideggerian Fourfold. At the opening of the autobiographical documentary devoted to Wright, his son appears on screen and murmurs a celebrated line by Ralph Waldo Emerson (1803–1882). I recognised in this a representation of Heidegger's architectonics, in the sense that representation theory within group theory gives to the word: one and the same abstract structure, the Fourfold, realises itself in distinct concrete spaces—Wright's organic architecture on the one hand, Emerson's transcendentalism on the other. The representation changes; the structure is preserved. Here again is an instance of what the present essay calls "structural resonance." I subsequently encountered the "aesthetics of atmospheres" (Ästhetik der Atmosphären) of Gernot Böhme (1937–2022) and the "new phenomenology"

(Neue Phänomenologie) of Hermann Schmitz (1928–2021), a philosophy of space as bodily lived. The "atmosphere" (Atmosphäre) that Böhme and Schmitz describe—a spatial quality sensed by the body prior to any division of subject and object—enters into direct resonance with the structure of the olfactory space that Corbin delineates in his history of sensibilities. For smell is experienced precisely as atmosphere: a diffuse quality pervading space, on this side of all articulation. When GI theory undertakes to constitute Corbin's olfactory space as a manifold, what that manifold aspires to describe may be nothing other than the mathematical counterpart of what Böhme and Schmitz called "atmosphere"—with the proviso, here as always, that what is at stake is not formalisation but structural resonance.

It was in the same period that I read the dromology of Paul Virilio (1932–2018). The "tyranny of speed" that Virilio describes—the acceleration of technical speed that transforms the structures of human perception and of politics—exhibits an unexpected point of contact with the "speed of structural change" measured by the Lie derivative in GI theory. In criticising "spatialised time," Bergson was objecting to the premise of a homogeneous flow of time. What Virilio denounces is the technical acceleration that compresses temporal experience to the point of rendering Bergson's "pure duration" impossible. The regions of the manifold where the Lie derivative takes abnormally high values—moments of extreme speed of structural change—are perhaps the very moments at which the "tyranny of speed" in Virilio's sense manifests itself upon the manifold.

To superimpose Heidegger's thinking on technology upon Virilio's dromology is to uncover a still deeper stratum. The Enframing (Gestell) that Heidegger describes in Die Frage nach der Technik (1953)—the essential structure of modern technology that challenges all beings forth as "standing-reserve" (Bestand)—is the obverse of Virilian acceleration. Under the regime of Enframing, nature and human beings alike are "set upon" (bestellt) as efficiently utilisable stock, and the being of beings is reduced to sheer availability. Virilio's acceleration is nothing but the temporal expression of this Enframing: to mobilise everything, ever faster. The regions of the manifold where the Lie derivative attains abnormally high values—moments of rapid structural transformation of the environment—are perhaps the moments at which the Enframing exerts its greatest violence, reorganising the ontological structure of the environment at forced march toward "availability."

The young Heidegger had wished to become a Catholic priest. Raised in a devout Catholic family in Meßkirch, he enrolled at the faculty of theology of the University of Freiburg im Breisgau. He subsequently turned to philosophy, yet the characteristic sensibility of Catholic spirituality—before the grasping of reason, listening; before the concept, reception—never ceased to run beneath the surface of his thought. The late Heidegger—the "flight of the gods" (Flucht der Götter), the awaiting of the advent of the divine ones, listening (Hören), that posture which consists solely in lending one's ear to the voice of Being—is less a matter of reason than of faith. Not to think but to feel; not to grasp but to receive. Gelassenheit—releasement, letting-be—: this concept, inherited from the mysticism of Meister Eckhart, is nothing other than the relinquishment of technical will, the receptive openness to the voice of Being.

Here I perceive a connection with the deepest Russian spirituality. The spiritual sensibility that dwelt in the simple peasants of the Russian earth—the one Dostoevsky depicted through the starets Zosima of The Brothers Karamazov: the kiss given to the earth, the forgiveness extended to all creatures—is a mode of openness to being radically distinct from Western rationalism. When the late Heidegger turned toward listening to the voice of Being, his thought communicated, beyond the language of reason, with the Russian spiritual tradition. And the "absolute nothingness" (zettai mu) of Nishida Kitarō—that predicative nothingness which pierces the very floor of subject-logic—is likewise a form of receptive, rather than grasping, cognition. The mu of Zen, Orthodox spirituality and Gelassenheit: three currents pointing in one and the same direction.

I perceived the connection between Russian spirituality, the world-view of the late Heidegger and the universe of the Kyōto School founded by Nishida, and this perception belongs to the first half of my youth. The Europeans and Americans among whom I grew up loved riding. Determined not to be thrown, I struggled with my small legs and my buttocks to absorb the horse's oscillations, and in that space which opened between the horse and my own body—that "in-between" (aida) which Kimura Bin was decades later to elevate into the central concept of his psychopathology—I encountered it for the very first time, without knowing what it was. Weizsäcker's Gestaltkreis—the cycle in which perception and movement guide and transform each other until they fuse inseparably—was already being realised in the body of the child on horseback. And in daily life, once I had dismounted, I learnt bodily, within the equestrian community itself, that the countries which eat horse meat and those which do not form two sharply divided camps, that for the latter the subject is taboo, and that, in short, the sensory frame of reference varies radically from one cultural sphere to another—that cultural structure of sensibility which Corbin was later to describe academically in his history of smell. In that equestrian community there also lived an elderly Japanese gentleman who carried on the cavalry tradition of the Imperial Japanese Army, in the line of Baron Nishi Takeichi—gold medallist in show jumping at the 1932 Los Angeles Olympic Games. Western equestrian culture and the Japanese tradition of the equestrian art crossed paths in one and the same riding school.

Orthodox Jews, too, lived near this community. Even when the sky suddenly turned to driving rain, they regarded the rain as a gift of nature and would not dream of opening an umbrella. A sacred book tucked under one arm, they trotted along at my side. In their silhouette, indifferent to the downpour, I glimpsed a form of the relation between nature and the human being that resembled neither the Japanese nor the European. What I learnt from them remains a precious possession.

The present essay lies in the prolongation of these primal experiences.

On the method of the present essay, a number of points must be clarified at the outset.

First, GI theory is not a mathematical formalisation of the existential analytic of Dasein. Martin Heidegger, in the seminars he conducted from 1959 to 1969 at Zollikon on the shore of Lake Zurich, in the home of the psychiatrist Medard Boss, expressly rejected any apprehension of human existence by the methods of the natural sciences. “All objectifying representations of a capsule-like psyche, a subject, a person, an ego or a consciousness in psychology and psychopathology must be abandoned”—such was Heidegger’s fundamental demand at Zollikon. The present essay respects this position without qualification. What it describes is the structural resonance (Strukturresonanz / résonance structurelle) between the phenomena that the existential analytic of Dasein describes in the language of words and those that GI theory describes in the language of mathematics. Neither translation, nor formalisation, nor reduction. Had Heidegger read this essay, he would in all probability have said: “This is not my work. But if there is something in yours that resonates with mine, it may mean that you have arrived, by a different path, at the same question concerning the structure of human existence.”

Second, Heidegger’s political career—the rectoral address of 1933 and his entanglement with National Socialism—lies beyond the horizon of the present essay. The literature on this question is vast; in the Japanese-language domain, the works of Kida Gen, Professor Emeritus at Chūō University, constitute the authoritative point of reference. The present essay confines itself to Heidegger’s dialogue with psychiatry in the framework of the Zollikon seminars.

Third, the relationship between the Kyōto School and the Imperial Japanese Navy—the problem of the colloquia known as “Japan and the Standpoint in World History” (1941–42)—is likewise not treated here. The intellectual heritage of the Kyōto School to which the present essay refers is limited to the logic of place of Nishida Kitarō and the psychopathology of Kimura Bin.

I should like to record here a word on the underlying motive of the present essay.

Kida Gen, looking back on the course of his own life, indicated that the defeat of 1945 had been the starting point of his philosophical quest. The social values that only the day before had been taken for granted were overturned to their foundations in the space of a single night—an event that truly deserves the name Kehre, and one that gave rise to an imperative need to seek, in the dimension of being, a ground capable of supporting the soil beneath one’s feet. This experience coincides exactly with that of Europe after the First World War: Spengler was read, and Der Untergang des Abendlandes (1918–1922); Husserl wrote Die Krisis der europäischen Wissenschaften (1936); Heidegger undertook to reinterpret from the ground up the history of Western philosophy since Socrates. When the ground of being trembles, there is no recourse but to direct one’s question at the structure of being itself.

The memory of Stefan Zweig (1881–1942) comes to mind. Die Welt von Gestern: Erinnerungen eines Europäers (1942). Born in Vienna at the twilight of the Habsburg monarchy, Zweig lived through the most luminous days of European civilisation in the 1920s—those days when, as he

believed, art, intellect and international fraternity had reached the highest point in the history of the human race. With the rise of Nazism, that civilisation collapsed. Zweig saw with his own eyes the horde of brownshirts leaping from lorries and flooding the streets; he became the helpless witness to a process in which human civilisation was regressing toward the Stone Age. After Kristallnacht (9 November 1938), he took the path of exile, a path that led him at last to Petrópolis in Brazil, where on 22 February 1942 he ended his life together with his wife Lotte. The post-war world, the nuclear arms race between the superpowers—these he was never to see.

The civilisational collapse Zweig described—the disappearance of the "world of yesterday" in the space of a single night—possesses, at the ontological level, the same structure as what Kida Gen experienced in the defeat of 1945 and what Husserl diagnosed in the Europe of the 1930s. In the language of GI theory, it is the moment at which scalar curvature plunges abruptly into the negative—the moment of the collapse of the structural stability of the manifold. Zweig could not survive that collapse. Heidegger drew from it an ontology. Kida Gen found in it the starting point of his philosophical quest. GI theory provides a mathematical instrument for describing the structure of such a collapse—though not its meaning: meaning lies beyond the reach of mathematics.

The meditation of Nishida and the Kyōto School was likewise inhabited by the will to elucidate the ontological structure of society and history. The "absolutely contradictory self-identity" and the "logic of place" were by no means abstract speculations: they were enterprises directed, from the place that is Japan and face to face with Western modernity, at interrogating anew the ground of being.

The approach of GI theory—differentiable manifolds, the Lie derivative, scalar curvature—differs radically from that of these philosophical traditions. Mathematics is not philosophy; the manifold is not ontology. In so far, however, as GI theory carries within itself the motivation to elucidate the structures of society, history and culture, it communicates underground with these traditions. How does the structure of an environment transform itself, collapse, reorganise? This question is Bergson's question about pure duration, Heidegger's question about temporality, Kimura's question about the "in-between"—and the question of the Lie derivative in GI theory.

These preliminaries established, the present essay poses the following question.

Bergson's pure duration, Heidegger's being-in-the-world (In-der-Welt-sein) and temporality (Zeitlichkeit), Kimura Bin's "in-between" (aida), Corbin's history of sensibilities—four intellectual traditions that have described, each in its own language, the temporal and spatial structure of human experience. GI theory possesses a mathematical framework capable of describing phenomena that stand in structural resonance with these descriptions. The present essay undertakes to lay bare the structure of this resonance.

Chapter One — Bergson's Pure Duration and the Lie Derivative

> *Il y a des changements, mais il n'y a pas, sous le changement, de choses qui changent : le changement n'a pas besoin d'un support.*
>
> "There are changes, but beneath the change there are no things which change: change does not need a support."
>
> 「変化はあるが、変化の下に変化する物はない。変化は支えを必要としない」
>
> „Es gibt Veränderungen, aber unter der Veränderung gibt es keine Dinge, die sich verändern: die Veränderung bedarf keines Trägers."
>
> *— Henri Bergson, L'Évolution créatrice (1907), Chapter IV*

In his Essai sur les données immédiates de la conscience, his doctoral dissertation of 1889, Henri Bergson (1859–1941) drew a decisive distinction between two conceptions of time. One is spatialised time—the time the clock marks, homogeneous, equidistant, countable. The other is pure duration (la durée pure)—the lived time in which qualitatively distinct moments interpenetrate and fuse within consciousness.

The core of the Bergsonian critique may be stated simply: the natural sciences reduce time to space. The time of the clock is nothing other than a sequence of homogeneous units—the second—arrayed along the number line: a spatial representation. The time of consciousness, however, is of an entirely different nature. The five minutes during which one finishes a cup of coffee differ qualitatively from the five minutes during which one awaits the result of an examination. To reduce this qualitative difference to the quantitative magnitude "five minutes" was, in Bergson's eyes, the fundamental error of modern science.

The Lie derivative of GI theory may be read as a mathematical response to this difficulty—without, it should be said, any pretence that the Lie derivative formalises "pure duration." Bergson would undoubtedly have declared that any mathematisation impairs the essence of duration.

The structural resonance lies at the following points.

The Lie derivative $\mathcal{L}_X g$ measures the extent to which the metric—that is, the way distances are measured—deforms along the temporal flow on the manifold. When $\mathcal{L}_X g = 0$, the metric is temporally invariant: the structure of space is "frozen." This is the mathematical counterpart of the "spatialised time" criticised by Bergson—a homogeneous time whose structure remains identical from instant to instant: a dead time.

When $\mathcal{L}_X g \neq 0$, by contrast, the metric deforms with time—the structure of space itself is "alive." Each instant possesses a qualitatively distinct structure. This enters into structural resonance with Bergson's pure duration—the succession of qualitatively irreducible moments.

In the case study devoted to the Ave Verum Corpus, the Lie derivative attained its maximum value—4.095—at bar 34. At that instant, the metric of the musical space underwent its most intense deformation: its structure differed qualitatively from that of the preceding bar and from that of the one which followed. In Bergson's language, this bar possesses the densest "duration"—not the repetition of a homogeneous time, but a qualitative change concentrated in a single instant.

Conversely, in the opening bars—"Ave verum corpus," bars 1 to 6—the Lie derivative was close to zero. The metric scarcely varied: the homophony in D major proceeded serenely. Yet this is not "dead time." It is perhaps what Bergson called "condensation of duration"—a silence pregnant with possibilities not yet unfolded. The Lie derivative of GI theory does not capture this nuance. This is, in all honesty, a limit of the method.

Chapter Two — Heidegger, the Zollikon Seminars and the Analytic of Dasein

> *Alle objektivierenden Vorstellungen einer kapselartigen Psyche, eines Subjekts, einer Person, eines Ich oder eines Bewusstseins in der Psychologie und Psychopathologie müssen zugunsten eines neuen Verständnisses aufgegeben werden.*
>
> "All objectifying representations of a capsule-like psyche, subject, person, ego or consciousness in psychology and psychopathology must be abandoned in favour of a new understanding."
>
> « Toutes les représentations objectivantes d'une psyché encapsulée, d'un sujet, d'une personne, d'un ego ou d'une conscience en psychologie et en psychopathologie doivent être abandonnées au profit d'une compréhension nouvelle. »
>
> *— Martin Heidegger, Zollikoner Seminare (1959), first session*

On 8 September 1959, in the lecture hall of the Burghölzli psychiatric university clinic in Zurich, Martin Heidegger (1889–1976) opened the first session with a concise exposition of the fundamental constitution of Dasein: being-in-the-world (In-der-Welt-sein). Some seventy psychiatrists sat in the audience, convened by Dr Medard Boss.

Boss had received a psychoanalytic training, but when, during the Second World War, serving as a military physician, he read Sein und Zeit, he found in it "a radically new, unheard-of insight into the mode of being of the human being in the world." In 1947 he visited Heidegger at his cabin in Todtnauberg; from that encounter grew an intellectual friendship that was to last a quarter of a century, together with the invitation to Heidegger to come and speak to psychiatrists at Boss's home in Zollikon, on the shore of Lake Zurich.

What did Heidegger wish to convey to the psychiatrists assembled at Zollikon? This, and nothing else: the conception of the human being presupposed by psychiatry—the separation of "mind" and "body" inherited from Cartesian dualism, the cleavage of subject and object, the partition between inner and outer world—rests upon a radical misunderstanding of the structure of human existence. The human being is not a subject locked within a psychic capsule, observing the world through a keyhole. The human being is through and through being-in-the-world: always already engaged in the world, open to the world, interwoven with the world. It is being-in-the-world that is the primordial unity—not the secondary joining of a subject and an object previously held apart.

The structural resonance with GI theory resides precisely in this structure.

On the manifold of GI theory, data points are never isolated observational values. Each data point is embedded in a "world"—the manifold itself—and its significance is determined by the surrounding metric structure. To wrench a data point from the manifold and analyse it in

isolation—this is exactly the procedure that Heidegger denounced under the name of the "capsule-like psyche." A single quarterly GDP figure, torn from the economic manifold, loses its meaning. That point begins to signify only when it is restored to the manifold—that is, when account is taken of the metric and the curvature that connect it to neighbouring points.

Yet let it be said once more: this is not a translation of Heideggerian ontology into GI theory. Heidegger's "world" is an ontological concept; the "manifold" of GI theory is a mathematical structure. What links them is not a conceptual correspondence but a structural resonance.

Chapter Three — Kimura Bin's "In-Between" and Geodesic Distance on the Manifold

> *There must exist on this earth something like the ground of life in general, and the fact that each of us is alive means that our existence maintains, on the plane of action as on that of sensation, a connection with this ground of life in general.*
>
> 「この地球上には、生命一般の根拠とでもいうべきものがあるはずであって、われわれ一人ひとりが生きているということは、われわれという存在が行為的にも感覚的にも、この生命一般の根拠とのつながりを維持しているということにほかならない」
>
> *— Kimura Bin, Aida [The In-Between], Kōbundō, 1988*

At the department of psychiatry in the Faculty of Medicine at Kyoto University, Kimura Bin (1931–2021) connected Nishida Kitarō's "logic of place" and "absolutely contradictory self-identity" with the clinical practice of psychopathology. A research stay at the University of Munich from 1961 to 1963, followed by one at the University of Heidelberg from 1969 to 1971, enabled him to absorb the phenomenological tradition of European psychiatry—Jaspers, Binswanger, Minkowski, von Gebsattel, Blankenburg—and to fuse it with the Japanese philosophical tradition, thereby founding a phenomenological psychopathology without parallel.

Its central concept is the "in-between" (aida). The "in-between" is neither a spatial gap nor a temporal interval: it is the relation itself between self and world, between self and other—the "place" of relation. Kimura conceived mental illness as a disorder of this "in-between": a fissure opens within it, a twist insinuates itself, or an opaque membrane interposes itself. Schizophrenia is the rupture of the in-between that links self and world; depression is its viscosity; epilepsy is its sudden conflagration.

Kimura further classified the temporal structure of mental illness by means of the metaphor of the festival. The ante festum—the temporality that anticipates a future not yet arrived—is schizophrenic. The post festum—the temporality that experiences an irretrievable lateness with respect to a past already gone—is melancholic. The intra festum—the temporality that immerses itself in the present instant—is epileptic.

Geodesic distance on the manifold of GI theory enters into structural resonance with Kimura's "in-between."

On a manifold, the distance between two points is measured not as a Euclidean straight line but as the length of the shortest path—the geodesic—along the curved surface. This distance is determined by the metric structure of the manifold—that is, by the way the space is curved. If the metric deforms ($\mathcal{L}_X g \neq 0$), the distance between the same two points changes: what was near may become far, and what was far, near.

The “collapse of the in-between” as Kimura described it—the rupture of the relation between self and world in schizophrenia—corresponds, in the language of GI theory, to an abrupt plunge of scalar curvature into the negative. When scalar curvature becomes strongly negative, the manifold assumes a saddle shape and geodesics diverge. The shortest path between two points lengthens abruptly, or ceases to be unique. The “distance” between self and world grows—not because the points have moved, but because the structure of the intervening space has collapsed.

In the case study of the Ave Verum Corpus, scalar curvature plunged to –490.72 at the passage “cuius latus perforatum” (bars 31–36). The “in-between” of the musical space was violently distorted at that instant: the body of Christ is pierced by the lance, and the structure of the musical space—not merely its content but its geometry—fractures. It is, in Kimura’s vocabulary, a passage where the “in-between” of the musical space collapses, and experience brushes against the schizophrenic temporality of the ante festum.

Yet Kimura’s “in-between” is the place of a living relation, borne by the vital ground-relation (Grundverhältnis in Weizsäcker’s sense). The geodesic distance of GI theory is no more than a geometric quantity. What connects them is, once again, a structural resonance, not a translation. Kimura would probably have turned away with a gentle smile from any attempt to reduce his “in-between” to a geometric quantity.

Chapter Four — Corbin's History of Sensibilities and the Manifold of Sensory Space

> *L'historien des sensibilités ne cherche pas à retrouver ce que les hommes du passé ont réellement senti ; il tente de comprendre le système de références qui organisait leurs émotions.*
>
> "The historian of sensibilities does not seek to recover what the people of the past actually felt; he attempts to understand the frame of reference that organised their emotions."
>
> *— After the method of Alain Corbin*

Alain Corbin's (1936–) Le Miasme et la Jonquille (1982) is the monumental work by which the third generation of the Annales school attained what has come to be called a "history of sensibilities" (histoire des sensibilités). Drawing on letters, diaries, decrees, medical publications and other qualitative sources, Corbin reconstructed the structural transformation of the olfactory space in France from the eighteenth to the nineteenth century: the passage from miasma theory to the hygienic revolution.

The point of contact between Corbin's methodology and GI theory has been discussed elsewhere (project note for the essay "Geometry of Sensibility"). I shall confine myself here to situating this tradition within the comparison of the four intellectual traditions that structure the present essay.

Corbin's history of sensibilities intersects with the traditions of Bergson, Heidegger and Kimura at the following points.

First, what Corbin describes is the historical change of the "sensory frame of reference." For the Parisian of a certain epoch, the smell of sewage signalled "danger"; the fragrance of flowers symbolised "virtue." This frame of reference was radically transformed by the construction of sewers and the advent of bacteriology. In the language of GI theory, this may be described as a change of the metric on the olfactory manifold—a non-zero Lie derivative. Just as Bergson's pure duration describes the qualitative change of consciousness, Corbin's history of sensibilities describes the qualitative change of the sensory "duration" of a society.

Second, Corbin's approach resonates with Kimura Bin's concept of the "in-between." Olfactory experience arises in the "in-between" of the smelling subject and the smelled object. What Corbin describes is the historical structural change of this "in-between": the same odour arouses terror in one epoch and passes unnoticed in another.

Third, Heidegger's concept of "being-in-the-world" casts fresh light on Corbin's olfactory space. Smell, unlike sight and hearing, is a sense for which it is difficult to maintain a distance from the object. Odour pervades space and blurs the boundary between subject and object. Olfactory experience is perhaps one of the most primordial forms of "being-in-the-world."

Chapter Five — Table of Structural Resonance

The table that follows maps the structural resonances between the four intellectual traditions and GI theory. Let it be said once more: these are resonances, not translations; correspondences, not identities.

Concept	Bergson	Heidegger/Boss	Kimura Bin	Corbin	GI Theory
Living time	Pure duration (la durée pure)	Temporality (Zeitlichkeit)	Time of the in-between ante/post festum	Historical change of sensibility	$\mathcal{L}_X g \neq 0$
Dead time	Spatialised time (temps spatialisé)	Vulgar concept of time (vulgärer Zeitbegriff)	Fixation of the in-between	Fixation of the frame of reference	$\mathcal{L}_X g = 0$
Structure of being	Stream of consciousness (flux de conscience)	Being-in-the-world (In-der-Welt-sein)	The in-between of self and world	Sensory frame of reference	Data points on the manifold
Structural collapse	Rupture of duration	Anxiety (Angst)	Depersonalisation, schizophrenia (collapse of the in-between)	Collapse of the sensory order on the eve of revolution	$R \to -\infty$ (negative scalar curvature)
Recovery / stability	Intuition (intuition)	Authentic existence (eigentliche Existenz)	Recovery of the in-between	Establishment of a new frame of reference	$R > 0$ (positive curvature)

The most important row of this table is "structural collapse." The rupture of duration in Bergson, anxiety in Heidegger, the collapse of the in-between in Kimura, the collapse of the sensory order on the eve of revolution in Corbin: all describe, in different languages, concerning different objects, a structurally identical phenomenon. The plunge of scalar curvature into the negative in GI theory is its mathematical counterpart.

It is, however, at the final row—"recovery"—that the limits of GI theory come to light. Bergson's "intuition," Heidegger's "authentic existence," Kimura's "recovery of the in-between": none of these designates a mere restoration of a prior structure. All imply a qualitative leap into a new dimension. The return of scalar curvature to positive values in GI theory registers the recovery of structural stability, but not the "qualitative leap." This limit must be reported in all honesty.

Final Chapter

> *絶対矛盾的自己同一*
>
> "Absolutely Contradictory Self-Identity" — Nishida Kitarō
>
> „Absolut widersprüchliche Selbstidentität“ — Nishida Kitarō
>
> « L’auto-identité absolument contradictoire » — Nishida Kitarō
>
> *— Nishida Kitarō, Zettai mujunteki jiko dōitsu [Absolutely Contradictory Self-Identity], 1939*

Nishida Kitarō’s “absolutely contradictory self-identity”—two things that contradict each other and that, in the very midst of their contradiction, are one—: this single concept distils everything the present essay has sought to say.

Consider Bergson’s pure duration alongside the Lie derivative of GI theory. They stand in flat contradiction: Bergson repudiates mathematisation, while GI theory is mathematics through and through. And yet, as though connected by a subterranean passage whose existence neither suspects, both bear witness to the same insight—that living time is time whose structure changes.

Or consider Heidegger’s being-in-the-world alongside the manifold of GI theory. The two are incompatible in their very mode of existence, the one an ontological concept, the other a mathematical structure. Nevertheless, each in its own idiom attests to a single truth: being cannot be grasped in detachment from the world.

Or again, set Kimura Bin’s “in-between” beside the geodesic distance of GI theory. They appear to belong to incommensurable orders, the first a living relation sustained by the vital ground-relation, the second no more than a geometric quantity. The conclusion that emerges from both, however, is the same: when the structure of relation collapses, the ground of being gives way beneath one’s feet.

Identical precisely in their contradiction—this is what Nishida meant by a “place.” The four intellectual traditions examined in this essay and GI theory diverge from one another, yet all of them, each speaking in its own voice, converge upon the same “place.” That place is nothing other than the dynamic itself whereby the temporal and spatial structure of human experience transforms, collapses and reconstitutes itself.

GI theory does not measure this “place.” What it does is describe mathematical phenomena that stand in structural resonance with it. And to perceive that resonance—is this not, when all is said, the most modest and at the same time the only honest form of what Bergson called “intuition”?

> *Die Sprache ist das Haus des Seins.*
>
> "Language is the house of being."
>
> 「言葉は存在の家である」
>
> « Le langage est la maison de l'être. »
>
> «Язык есть дом бытия.»
>
> *— Martin Heidegger, Brief über den Humanismus (1947)*

If language is the house of being, then mathematics is perhaps another house of being. Not the same house, mark you. Another. And both stand upon the same ground.

But whether these two houses will remain forever distinct is something the mathematics of the twenty-first century may settle.

In contemporary mathematics, the Langlands programme is the enterprise that seeks to unify, by means of a vast network of correspondences, domains long believed to be unrelated: number theory, algebraic geometry, representation theory, harmonic analysis. Born of a letter that Robert Langlands addressed to André Weil in January 1967, this programme has formed the vanguard of mathematics for more than half a century, and through the geometric Langlands correspondences now extends its reach to physics—quantum field theory and string theory.

When all the pieces are in place, mathematics and the natural sciences may perhaps return to the form of knowledge that obtained before the Renaissance. In the medieval universities of Europe, the seven liberal arts (septem artes liberales)—the trivium (grammar, rhetoric, dialectic) and the quadrivium (arithmetic, geometry, astronomy, music)—were taught as an organic whole. Music was a branch of mathematics, astronomy an application of geometry, dialectic the method that traversed them all. Natural science, mathematics, philosophy, the fine arts, morals: not walled-off disciplines, but the diverse aspects of a knowledge laid out upon a single map. The modern division of intellectual labour tore that map to pieces; the advances of the Langlands programme suggest the existence of a thread capable of stitching the fragments back together.

Werner Marx, heir to the late Heideggerian thought, posed in Gibt es auf Erden ein Maß? Grundbestimmungen einer nichtmetaphysischen Ethik the question of the possibility of an ontological ethics. The question of the "measure" that Heidegger had raised in meditating upon the line of Hölderlin (»...dichterisch wohnet der Mensch...«), Marx unfolded as the fundamental question of ethics. And to set a measure—to decide what is good, what is beautiful, what is true—resonates structurally with the act of imposing a metric on a manifold.

Categorical logic, for its part, is elucidating the logical rules that hold within each categorical structure and the passages of logical translation between categories—translatability by functors. An "in-between" exists between different mathematical worlds, and the functor that traverses this "in-between" translates the theorems of one world into the theorems of the other. Just as Kimura Bin's "in-between" was the place of the relation between self and world, the functor is the place of the relation between one mathematical world and another.

In this context, one body of work commands my attention: the "Categorical Unified Science" proposed by Maruyama Yoshihiro (Hakubi Centre and Faculty of Letters, Kyoto University / Department of Computer Science, University of Oxford; currently Australian National University). Conceiving of category theory as a "theory of structural networks," Maruyama showed that the deductive networks woven by propositions, the causal networks woven by physical systems, the semantic networks woven by words and sentences, the social networks woven by agents—all of these are merely diverse instances of one and the same structural network called a category. What is decisively important in Maruyama's project is that it does not amount to the reductionist unified science of logical positivism, but to a pluralistic unification that does not flatten the inherent complexity of each science. "Unity as the networking of knowledge, not as reduction or uniformisation": this ideal resonates profoundly with the methodological stance of the present essay, which seeks between GI theory and Bergson-Heidegger-Kimura not a formalisation but a structural resonance.

As Maruyama himself stresses, the "predicative logicism" of Nishida Kitarō—the logic of the predicate (logic of place) as opposed to the logic of the subject (Aristotelian logic)—is analogous to the process philosophy that abandons the logic of substance in order to found itself upon becoming (Becoming) rather than being (Being); categorical unified science connects naturally with the philosophy of the Kyōto School. Maruyama further points to the continuity with Nishi Amane's "Hyakugaku renkan" (Encyclopaedia of the Sciences)—a pioneering project of unified science in Japan—and with Nishitani Keiji of the Kyōto School, who proclaimed the urgency of "constructing a unified world-view" in a "nihilistic age of the fragmentation of world-views."

Separately, Ōyama Tomonori, in "The Development of Social Theory by Means of Category Theory," employed the categorical adjunction as an instrument for the unified translation of Hashizume Daizaburō's language-game theory, Imada Takatoshi's self-organisation, Satō Toshiki's voluntaristic theory of action, Lacan's Schema L and Benjamin's critique of violence, and obtained the remarkable result that social theories which, in natural language, present entirely different faces bear a striking resemblance to one another within the structure of the categorical adjunction. Ōyama's work constitutes an effective demonstration that Maruyama's programme is applicable to concrete questions of theoretical sociology, and it communicates methodologically with GI theory's attempt, in its quantitative social science paper, to establish "structural correspondences with existing methods."

GI theory addresses the same question by a different route from Maruyama's categorical unified science: through the Riemannian manifold and the Lie derivative. Where Maruyama founds his

approach on the algebraic structure of category theory, GI theory founds its own on the analytic structure of differential geometry. Yet both aim at the same horizon: a networking of knowledge that transcends the fragmentation of the sciences. Whether the two will one day converge—at the meeting point of category theory and differential geometry, precisely the domain in which derived categories and the theory of motives dwell—is one of the questions to which twenty-first-century mathematics must give an answer.

Shinichi Mochizuki, a mathematician currently at Kyoto University—the intellectual heir of Nishida Kitarō—has proposed inter-universal Teichmüller theory (IUT), a solitary theoretical edifice whose verification by the mathematical community at large remains incomplete. This attempt to throw bridges across the "in-between" of several mathematical universes, beyond the framework of the conventional mathematical universe—an attempt which appears in itself to be a mathematical enactment of Nishida's "absolutely contradictory self-identity"—will it flow into the movement of the Langlands programme, that intellectual grand unification in progress?

We may learn the answer in the course of the twenty-first century.

Or we may not. Just as Zweig never saw the post-war world. But in the very fact of having posed the question—in the fact that Bergson interrogated duration, that Heidegger interrogated being, that Nishida interrogated place, that Kimura interrogated the "in-between," that Corbin interrogated sensibility, that Langlands interrogated correspondence, that Mochizuki interrogated the inter-universal—in that fact itself resides the dignity of the human intellectual enterprise.

The mathematical unification of general relativity and quantum mechanics—at a stage still prior to quantum field theory—was, in all likelihood among the very first in the world, calculated by the pen of a young Russian born in the Soviet Union, gifted in the humanities and the arts no less than in the sciences. Matvei Petrovich Bronshtein (Матвей Петрович Бронштейн, 1906–1938) conceived the cGħ cube and carried out pioneering research on quantum gravity. A gifted author of science books for children, he departed this world at the age of thirty-one. His name I shall not forget. The earth and the sky of Russia keep his memory.

I perceive in each language a timbre that belongs to it alone, a beauty no other possesses. When I decide in which language to express—or re-express—this text, I obey the sensory desire to know in which timbre I wish to savour it.

It deserves mention that Japan's encounter with Western civilisation did not begin at the close of the Edo period but reaches back to the age of Azuchi-Momoyama, when Christian missionaries, members of the Society of Jesus foremost among them, arrived one after another in the domains of the various daimyō. What is striking about this encounter is that the political elite of the day—

the daimyō and their retainers—no less than the urban and rural populace, received these guests not passively but on their own initiative entered into cultural, civilisational and, in the full sense of the word, theological dialogue with them. Roman Catholics and Buddhists, the latter chiefly Zen monks who took pleasure in academic disputation, debated Catholic theology by means of Zen logic, motivated not by religious prejudice nor by any reflex of exclusion, but by a purely intellectual desire for truth. Your argument contains a contradiction: which of the two theses is the true one?—such were the terms in which the disputatio was conducted. Not a clash of dogmas but a collision of logically constructed arguments, a chain of refutations the like of which would be hard to find elsewhere. The fathers and missionaries, reporting on these encounters in letters dispatched to the headquarters of the Society and to the Holy See, bore testimony to the following effect: to convert the Oriental people of this land to Christianity is more difficult than in any other country, for they reply at the level of the intellect—logically, rationally. The towns are clean, the level of education among the people remarkably high, their conduct restrained, and they live in a spirit of mutual cooperation. The daimyō are, so to speak, kings of small or medium kingdoms, and some rule over large domains. And in Kyōto, as a religious authority, there resides an emperor—not a King, but a figure who ought to be called Emperor.

In that epoch, the encounter between Japan and the Western world had not yet taken the form of intimidation by force of arms; it was an intellectual exchange, and one of the most captivating in the annals of world history. Were this period to serve as the subject of an empirical historiography founded upon manifolds, what results would the Lie derivative, the Lie bracket and the remaining instruments of geometric structural analysis bring to light? The historical sources, both numerical and qualitative, appear to have been preserved in Japan in considerable volume.

It is my hope that the Japanological traditions of France, Britain and the United States, heirs to a rich heritage of Oriental scholarship, will turn their attention to GI-theory-based empirical historiography. To know the past of a nation is to penetrate to the depths of the principles of action and the unconscious strata of consciousness of the Japanese of today, and such knowledge ought to be of no small value for the political practice of those Western nations themselves. Japan, for its part, must analyse the past of each of these countries in order to understand them better, while at the same time sharpening its ear for the manner in which it is itself being analysed. Such is the intellectual contest of our age—a voiceless contest, a silent game of strategy in which each party takes care that the adversary shall not hear what resolution it has adopted. The noblest of games between gentlemen.

Amen.

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